Mikhail Razakov

Quantitative Research Candidate — Python · C++ · Time-Series Research

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PROFILE

I work on algorithmic research and reproducible experiments in Python and C++. I prepared an XAGUSD OHLC benchmark and a blind price-channel calibration interface. My other projects compare verification policies using frozen fault cases, valid controls and reference oracles. I work on symbolic computation at ISP RAS and previously completed a C++/LLVM compiler internship at MCST. Predictive financial effectiveness is not established.

TECHNICAL SKILLS

Languages and data: Python, NumPy, pandas, SciPy, C++23, GNU C17, SQL, Git, Linux

Mathematics and algorithms: Integer arithmetic, affine models, modular/GCD checks, graph algorithms, OHLC and time-series processing

Research validation: Deterministic experiments, valid controls, reference oracles, blind calibration, randomized/metamorphic testing and counterexample analysis

EXPERIENCE

ISP RAS · SharpChecker · October 2026 — present

• Working on symbolic computation for C#/.NET static program analysis.

MCST · Compiler Engineering Internship · July — August 2026

• Worked on experimental LLVM LICM with Python before/after checks. Implemented RPO, Dijkstra, Dinic and Tarjan SCC; a local fixture run passed 14/14 tests.

ENGINEERING PROJECTS

Simple Parallel Channel Tracker

Built a private Python multi-timeframe OHLC research prototype and a deterministic benchmark of 31 heuristically labeled cases across 10 timeframes. Both structural validators passed; blind calibration has 20 cases and eight relations. Labels were constructed retrospectively; there is no expert gold set, validated prediction or established profitability. Source remains private.

UniversalToolchain / Wist2

Built an experimental comparison of four policies with frozen author-designed faults, valid controls, additional post-freeze cases and independent oracles. The outcomes are reproducible within the UniversalToolchain study, but internally designed corpora limit external validity. This is compiler-systems research, not a financial-market experiment.

PS-form Memory Dependence Analyzer

Built a GNU C17 analyzer using exact counting for supported affine cases, modular/GCD reasoning and comparisons to an independent oracle. An extended test run found five conservative maybe results where the oracle expected no. This reflects analysis incompleteness, not a financial forecast.

EDUCATION

HSE University — Software Engineering

RECOGNITION

2025–2026 · MEPhI Junior: First-degree diplomas in the Information Technologies section in 2025 and 2026.

2026 · MCST LLVM/C++ selection: According to organizer correspondence: 5.0/5.0 and 104/104 official tests among 49 submitted solutions.

2026 · LangDev — UniversalToolchain: Presented on 9 October 2026; organizer published the talk recording. — Talk recording